RALIO TECHNOLOGIES

FEBRUARY 2025

Mercury Microcontroller Development Board - Version 2

Datasheet Revision 1.0

www.raliotech.com

Mercury

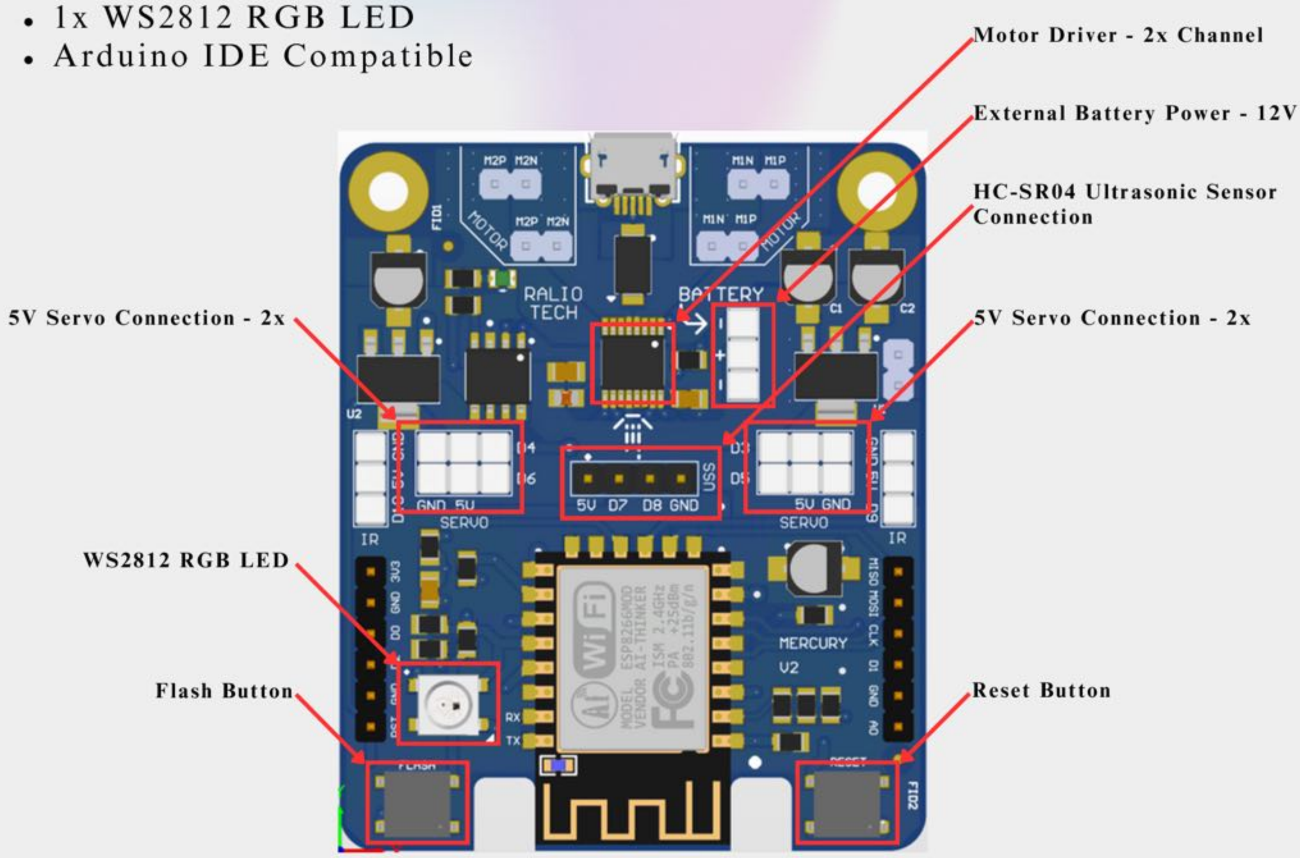
V2

Introduction

Mercury Development Board is a versatile IoT platform comprising of onboard Wi-Fi and motor driver support. It is developed using ESP-12 module, which has ESP8266 Wi-Fi SoC from Espressif Systems.

Key Features

- 3.3V operated
- Can be powered through USB
- Can be powered externally up to 12V
- Uses wireless protocol 802.11b/g/n
- Capable of PWM, I2C, SPI, UART
- 1x Analog pin
- 2x Onboard Motor driver channel (1A max. per channel)
- 4x Onboard direct 5V servo connections
- 1x HC-SR04 Ultrasonic sensor connection
- 1x WS2812 RGB LED
- Arduino IDE Compatible



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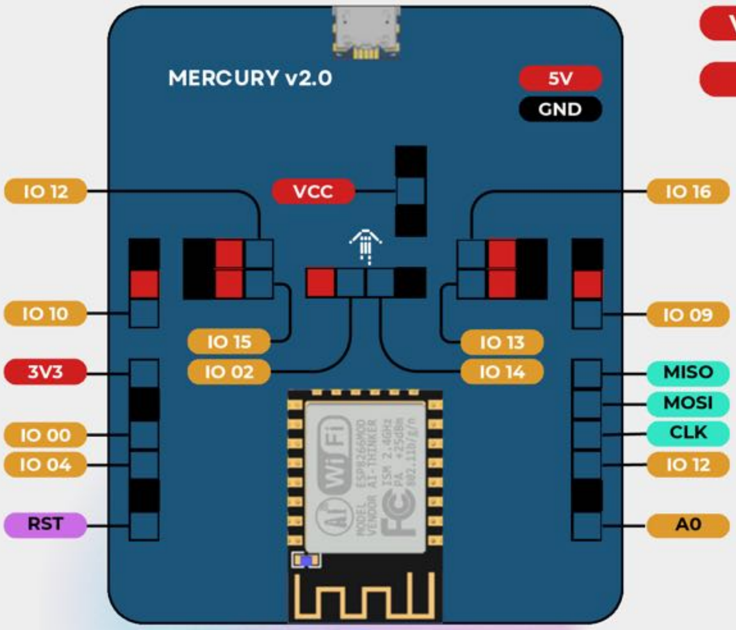
Pinout Diagram



VCC 12V Maximum

5V 1A Maximum

Pin	GPIO	Onboard Usage (1)	Onboard Usage (2)
D0	GPIO 0	WS2812 RGB LED	
D1	GPIO 12	MOTOR 1 PWM	
D2	GPIO 4	MOTOR 2 PWM	
D3	GPIO 16	MOTOR 1 DIRECTION	SERVO MOTOR
D4	GPIO 5	MOTOR 2 DIRECTION	SERVO MOTOR
D5	GPIO 13		SERVO MOTOR
D6	GPIO 15		SERVO MOTOR
D7	GPIO 2	BUILTIN LED	ULTRASONIC SENSOR TRIGGER
D8	GPIO 14		ULTRASONIC SENSOR ECHO
D9	GPIO 9		
D10	GPIO 10		

Mercury

V2

Technical Specification

Board	
Name	Mercury
Version	2
SKU	100008
Microcontroller	
Name	ESP8266EX
Peripheral Interface	
Digital I/O	11
Analog Input	A0 (0 - 1V)
PWM (Hardware)	GPIO 4, GPIO 12, GPIO 14, GPIO 15
Connectivity	
USB	Micro USB
WiFi	2.4 GHz
Communication	
UART	Yes
SPI	Yes
I2C	Yes

Mercury

V2

Technical Specification

Power	
I/O Voltage	3.3V
External Input Voltage (VIN)	7V - 9V (nominal)
DC Current per I/O pin	10mA
Voltage Rails	3.3V; 5V; VIN
Battery connector	3 Pin Male Header (1: Ground, 2: Power, 3: Ground)
Onboard Peripheral Support	
Builtin LED	Yes
RGB LED	Yes
Motor Driver	2 Channels; 1A maximum per channel;
Servo Connection	Yes; 4x
HC SR04 Connection	Yes
Dimension	
Length	59mm
Width	48mm
Height	12mm
Color	Blue

Mercury

V2

Arduino® IDE Support

Detailed instructions can be found on this Github page:
https://github.com/raliotech/products/tree/master/boards/mercury

Board JSON File:
https://raw.githubusercontent.com/raliotech/engineering/master/core/package_ralio_index.json

Code Flashing

After successful code verification, click "Upload" button on Arduino IDE. **Press and Hold** "Flash" Button on the board when you see "Connecting..." in Arduino console. Let go off the button once the code starts to upload.

Reference

https://www.espressif.com/sites/default/files/documentation/0a-esp8266ex_datasheet_en.pdf

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